Prediction of Convection Zone Depth in Marine Tidal Structures using Symbolic Regression

Prediction of Convection Zone Depth in Marine Tidal Structures using Symbolic Regression

Report submitted to

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In partial fulfilment of the requirements for the award of the degree

of

Master of Technology in Structural Engineering

by

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Under the supervision of

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SCHOOL OF INFRASTRUCTURE

INDIAN INSTITUTE OF TECHNOLOGY BHUBANESWAR

May,2026

Dedicated

To

My Family, Faculty & Friends

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DECLARATION

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c) I have followed the guidelines provided by the Institute in preparing the thesis.

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Aastha Pangaria

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Date: **Aastha Pangaria**

DECLARATION

I, Aastha Pangaria, hereby declare that the work presented in this thesis, entitled **"**Prediction of Convection Zone Depth in Marine Tidal Structures using Symbolic Regression**"**, is an authentic record of my own research work carried out under the supervision of Dr. Padala Suresh Kumar at the School of Infrastructure, Indian Institute of Technology Bhubaneswar.

This work was carried out in partial fulfilment of the requirements for the degree of Master of Technology in Structural Engineering.

I further declare that the matter embodied in this thesis has not been submitted by me for the award of any other degree or diploma in this or any other university or institute.

Signature of the Student

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ABSTRACT

The service life of reinforced concrete in seawater is primarily governed by chlorides and water penetration, causing corrosion of the embedded steel. The key phenomenon in this process is the development of the convection zone, a surface zone where cyclic wetting and drying result in chloride accumulation in depths from the surface. The determination of the depth of the convection zone (CZD) is crucial for service-life prediction of marine structures.

This research investigated the convection zone depth in terms of a blend of civil engineering concepts and data-driven modeling. Experimental findings reported in the literature, such as chloride profiles and moisture transport indices, were examined to detect the effect of exposure conditions and material characteristics. Symbolic Regression (SR), which was performed through the PySR environment, was utilized to identify explicit mathematical relationships between CZD and governing parameters like dry/wet ratio.

While PySR provided highly accurate baseline equations for individual ports, it struggled to generalize across the global dataset due to discrete, systematic variations in the FEM mesh data. To address this, targeted parameter isolation was performed. By keeping the cycle times mathematically constant, the analysis proved that environmental weather intensity has a negligible impact on penetration depth. A Random Forest machine learning model was deployed to handle these complex data variations. The extracted feature importance values quantitatively indicated that the physical cycle durations strictly dictate the depth of the convection zone, with drying time (~51%) and wetting time (~38%) driving nearly 90% of the transport mechanism.

Keywords: Convection zone; Marine concrete; Moisture transport; Symbolic Regression; PySR; Equation discovery; Durability modeling

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LIST OF ABBREVIATIONS AND NOTATIONS

Abbreviations

ANN	Artificial Neural Network
CWAA	Cyclic Water Absorption Amount
CZD	Convection Zone Depth
DWTR	Drying to Wetting Time ratio
GP	Genetic Programming
IDMT	Influential Depth of Moisture Transport
MAE	Mean Absolute Error
PCC	Peak Chloride Concentration
PySR	Pythonic Symbolic Regression
SR	Symbolic Regression
SVM	Support Vector Machine
WDA	Wetting-Drying Cycles

Notations

t_{wet}	Duration of the wetting period in a cycle(seconds)
t_{dry}	Duration of the drying period in a cycle(seconds)
R2	Coefficient of Determination
μ	Mean of a feature's values
σ	Standard deviation of a feature's values
y_i	The actual, observed value of the target variable for the i-th data point
$\hat{y}_i$	The model's predicted value of the target variable for the i-th data point
$\bar{y}$	The mean of all observed values of the target variable
n	The total number of data points in a set

Chapter 1
Introduction

General

This chapter sets the stage for the current study. The emphasis is placed on simulating moisture transport in concrete structures exposed to sea environments. The chapter begins by describing the general durability issues these structures experience, with a particular emphasis on the fundamental role played by moisture dynamics. It then presents the primary physical phenomena at play. This involves the creation of a near-surface "Convection Zone," which is primarily governed by wetting and drying cycles. The text also sets forth the key indices to describe this behavior: the Convection Zone Depth (CZD). Lastly, the chapter establishes the scope of research and sets forth an outline of the thesis structure.

1.1 The Durability Challenge in Marine Environments

Reinforced concrete is a common material for coastal and marine structures, e.g., bridges, ports, and offshore platforms. It is selected on the grounds of strength and economics. The service life of such structures usually relies more on durability than on mechanical strength. They are required to be able to handle extreme environmental loads. The sea environment is very aggressive. It exposes concrete to a continuous barrage of physical and chemical attacks. Physical attacks involve wave action and abrasion, whereas chemical attacks arise from moisture and sea salts.cm

Transport of moisture is a leading cause of this breakdown. Moisture penetrates the porous structure of concrete and serves as a carrier for aggressive substances, e.g., seawater chlorides. The high alkalinity of concrete ordinarily develops a passive, protective coating around embedded steel reinforcement. Nevertheless, the entry of these agents can degrade this film. The degradation activates an electrochemical process of corrosion. As the steel corrodes, rust (iron oxides) is formed, and it occupies a volume several times that of the original steel. This growth produces tremendous internal tensile stresses within the concrete, which is itself a weak tensile material.

Ultimately, these internal stresses produce cracking and "spalling." Spalling happens when substantial portions of the concrete cover spall off the surface. Figure 1.1 presents a graphic illustration of this failure mode. It displays a concrete column where the cover has fully spalled off, revealing the highly corroded steel reinforcement cage. This deterioration weakens the steel-concrete bond. It also lowers the effective cross-section of the structural member. Such problems considerably impact the structure's load-carrying capacity and safety. The economic implications are enormous and include expensive cycles of inspection, rehabilitation, and repair. Thus, a predictive understanding of the underlying moisture transport is a cornerstone of service-life prediction and durability-based design for all coastal structures.

Figure 1.1 Severe spalling and corrosion at the base of a reinforced concrete column

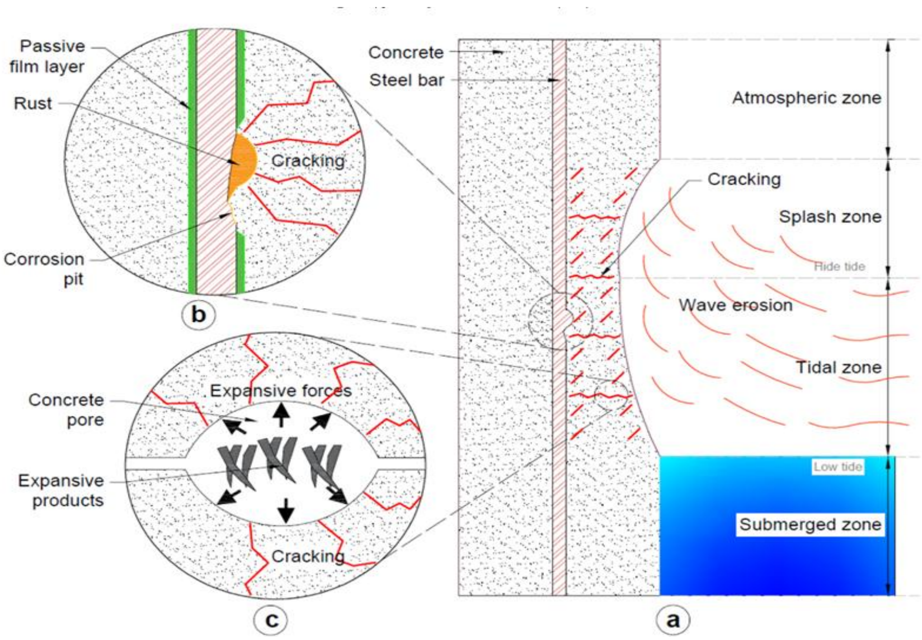

Figure 1.2 Concrete exposed to deterioration in marine environment: (a) Element of concrete structure, (b) steel corrosion and (c) damage by expansion (Mehta and Monteiro, 2006)

1.2. The Convection Zone

For many years, engineering models have relied on simplified assumptions to predict the ingress of harmful substances. One such assumption is Fickian diffusion. This approach assumes transport is driven purely by a concentration gradient into a saturated, uniform material. As illustrated in **Figure 1.2(a)**, a marine structure is exposed to several distinct microclimates. These include the submerged, tidal, splash, and atmospheric zones. A diffusion-based model is a reasonable simplification for constantly submerged structures. However, it completely fails to capture the dominant transport physics in the most aggressive regions: the tidal and splash zones.

These zones are subjected to relentless cyclic wetting and drying. This process induces a powerful transport mechanism that overrides pure diffusion in the near-surface region. During the wetting phase of a cycle (defined by duration t_{wet}), such as a high tide or wave splash, the unsaturated pores of the concrete absorb moisture through strong capillary suction. During the drying phase (duration t_{dry}), moisture evaporates from the surface. This leaves any dissolved substances behind to accumulate within the pore structure, a process governed by the varying Drying Flux (q_{dry}). This repeated cyclic transport process creates a distinct near-surface layer known as the convection zone, where the internal

Equilibrium Moisture ($\theta_{r,\,eq}$) fluctuates significantly. As shown in Figure 1.2(b, c), this process is the direct cause of localized corrosion and the subsequent internal cracking from expansive rust products.

This zone's behavior is most accurately described by two important indices: the Convection Zone Depth (CZD). The CZD specifies the depth to which such cyclic moisture variations extend. This furnishes a direct quantitative measure of the material's susceptibility to its environment. The existence of this convective transport of moisture renders the fundamental premises of Fick's law inapplicable. The structure of Fick's law is not capable of accommodating these elaborate, circular dynamics.

1.3 Scope of the Study

The scope of this study is centred on the development of an interpretable data-driven framework for analyzing the Convection Zone Depth (CZD) in marine concrete structures subjected to cyclic wetting and drying exposure conditions. The research utilizes a large Finite Element Modelling (FEM)-generated dataset representing a wide range of environmental exposure scenarios and moisture transport conditions.

Unlike conventional approaches that rely solely on empirical fitting or black-box prediction, the present study focuses on identifying the dominant physical transport mechanisms governing CZD evolution. Four coupled environmental variables were initially considered to characterize the marine exposure process:

1. Wetting Duration (t_{wet})
2. Drying Duration (t_{dry})
3. Drying Flux (q_{dry})
4. Equilibrium Moisture ($\theta_{r,eq}$)

The study applies Symbolic Regression (PySR) to identify physically interpretable mathematical relationships between the governing environmental variables and CZD evolution. Based on the recurring symbolic patterns obtained from the initial analysis, a reduced-order normalized transport formulation was developed using dominant temporal exposure variables.

To further investigate generalized transport behavior, coefficient extraction and Drying-to-Wetting Time Ratio (DWTR)-based analyses were performed to establish universal coefficient relationships governing drying-controlled and wetting-controlled transport mechanisms. Random Forest validation was additionally employed to independently evaluate the relative influence of the environmental variables and validate the dominant role of cyclic exposure durations.

The research does not involve new experimental testing; instead, it focuses on extracting physically meaningful reduced-order transport relationships from the FEM-generated dataset and establishing a generalized analytical framework capable of representing moisture transport evolution across varying marine exposure conditions.

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1.4 Thesis Outline

This thesis is structured into five chapters to present the research methodology, analysis procedures, and findings in a systematic and logical manner. Chapter 1 introduces the problem of moisture transport in marine concrete structures, discusses the concept of the convection zone, and defines the scope, objectives, and motivation of the present study. Chapter 2 presents a comprehensive review of the existing literature related to moisture transport mechanisms, convection zone development, cyclic wetting–drying exposure behavior, data-driven modelling approaches, and Symbolic Regression techniques relevant to marine durability analysis. Chapter 3 describes the complete research methodology adopted in this study. This chapter includes details regarding the FEM-generated dataset, preprocessing procedures, feature engineering, Symbolic Regression implementation, coefficient extraction methods, DWTR formulation, Random Forest validation framework, and the overall analytical workflow used throughout the study. Chapter 4 presents and discusses the results obtained from the multi-phase analytical framework. The chapter includes the symbolic regression exploration, development of reduced-order transport equations, coefficient extraction procedures, DWTR-based coefficient analysis, universal coefficient function development, Random Forest-based validation of dominant transport mechanisms, and discussion of model strengths and limitations. Chapter 5 summarizes the major conclusions obtained from the study, highlights the primary physical insights regarding convection-zone transport behavior, discusses the engineering significance of the proposed framework, and outlines recommendations for future research and further model development.

Chapter 2
Literature Review

Building on the durability challenges discussed in the previous chapter, this review examines the scientific literature to provide a detailed theoretical foundation for the present study. The primary goal is to critically assess the current understanding of moisture transport in concrete exposed to marine environments and to evaluate the existing tools for modeling these processes.

The chapter begins by describing the fundamental transport mechanisms, with a focus on the physics of wetting-drying cycles and the formation of the convection zone. These processes are quantified using key indices, particularly the Convection Zone Depth.

Following this, a systematic review of established modeling approaches is presented, ranging from simple empirical formulas to complex numerical simulations. The strengths and, more importantly, the limitations of each approach in capturing moisture-driven transport are discussed. The review then turns to modern data-driven techniques, contrasting opaque "black-box" models with the more transparent approach of Symbolic Regression (SR).

The chapter concludes by highlighting the research gap that remains in this field. This gap directly motivates the present study and defines its objectives.

2.1 Transport Phenomena in Marine-Exposed Concrete

The long-term durability of concrete used in coastal and offshore structures is governed by transport processes within its porous microstructure. Among the various agents of degradation, the cyclic movement of moisture is the primary driver in the most aggressive marine exposure zones. Moisture movement enables the ingress of deleterious substances such as salts, a process driven by the environmental Flux (q_{dry}), and accelerates the deterioration of the concrete. Therefore, understanding how moisture is transported in marine conditions is essential for predicting service life and improving durability.

2.1.1 The Dominance of Wetting-Drying Cycles

Concrete in marine environments is exposed to different microclimates, including submerged, tidal, splash, and atmospheric zones. In the permanently submerged zone, transport is slow and mainly controlled by diffusion into a saturated medium.

In contrast, the most severe degradation occurs in the tidal and splash zones. These regions are subjected to relentless wetting-drying cycles (WDA), which drive a strong cyclic transport mechanism that is far more aggressive than simple diffusion. During wetting periods, such as high tide or wave splashes, unsaturated concrete absorbs water due to strong capillary suction. During drying periods, evaporation removes water from the near-surface pores. This continuous ingress and removal of moisture not only stress the concrete's microstructure but also acts as the primary transport pathway for dissolved substances, including sea salts and chlorides. The critical influence of these cycles has been extensively documented. Medeiros et al. (2013) showed that cyclic exposure results in a significantly higher ingress of aggressive agents compared to permanently submerged conditions.

2.1.2 The Convection Zone

The repeated pumping action caused by WDA creates a distinct near-surface region known as the convection zone, often referred to as the "skin". This zone experiences strong fluctuations in internal Equilibrium Moisture ($\theta_{r,\,eq}$) and is the focal point of durability degradation.

The characteristics of this zone are best described by two important moisture-transport indices. The first is the Convection Zone Depth (CZD), which defines the depth from the concrete surface affected by significant cyclic moisture variations. It marks the physical boundary of the convection zone's influence.

The second is the Cyclic Water Absorption Amount (CWAA), which quantifies the net volume of water absorbed and released by the concrete during a single wetting-drying cycle. This index reflects the "breathing" capacity of the concrete and is directly related to the intensity of the transport mechanism.

Experimental studies by Zhang et al. (2018) and modeling work by Padala et al. (2022) have established a direct relationship between these moisture indices and the formation of the convection zone. They showed that CZD determines the range of the convection zone, while CWAA correlates with the amount of substance accumulated within it.

The parameters of the wetting and drying cycles are particularly important. Research by Cao et al. (2022) demonstrated that the ratio of drying time to wetting time directly controls the depth of the convection zone (CZD). Longer drying periods lead to a more pronounced and deeper convection zone. However, these studies often overlook the varying intensity of Flux during these cycles, which significantly modulates the severity of accumulation.

Moisture-driven accumulation also shapes the concentration profiles of dissolved solutes such as chlorides. These profiles typically exhibit a peak concentration at a certain depth below the surface, a phenomenon that cannot be explained by simple diffusion. This peak is described by the Peak Chloride Concentration (PCC) and the Depth of Peak Chloride Concentration (DPCC). The DPCC is essentially synonymous with the depth of the convection zone and is therefore directly linked to the CZD.

Field studies by Ting et al. (2021) and a comprehensive review by Cai et al. (2020) confirmed that WDA can expand the convection zone by a factor of two to three compared to submerged conditions. These observations highlight the limitations of inadequacy of traditional diffusion-based models in capturing the transport behavior under wetting-drying exposure.

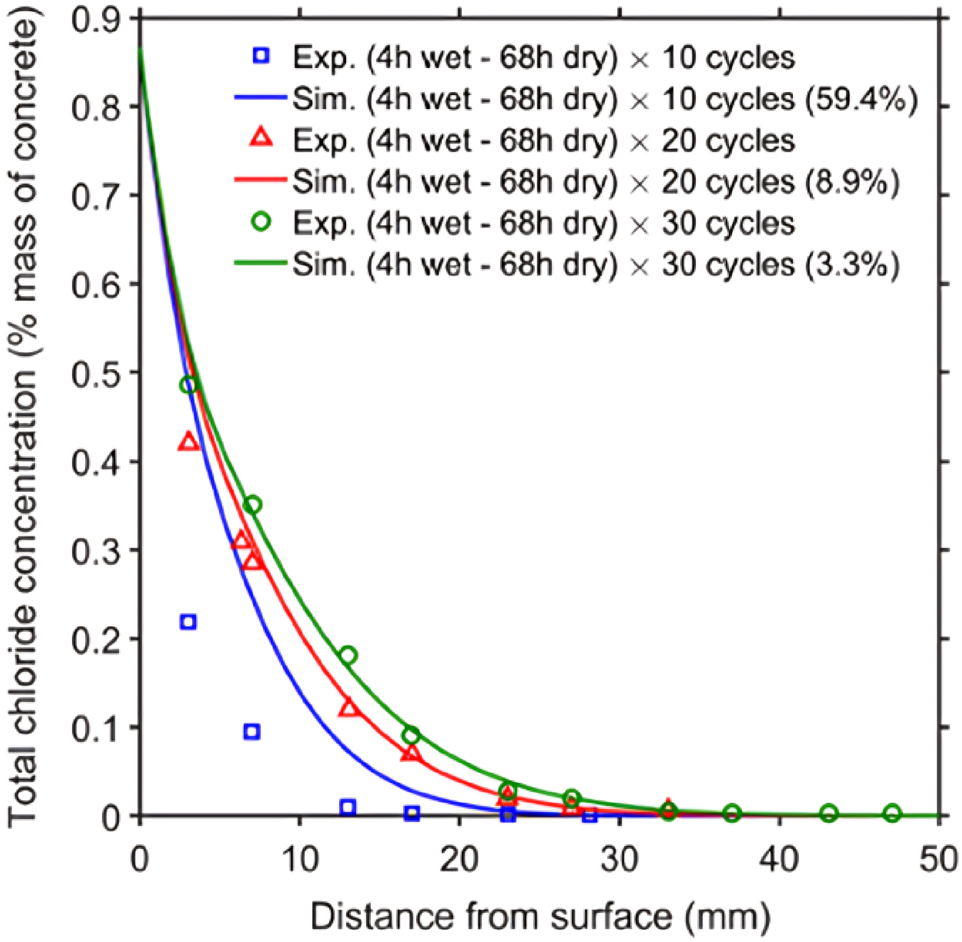

Figure 2.1 Typical chloride concentration profiles in concrete under wetting-drying cycles; results from Chen et al. (2021)

The existence of the convection zone has been widely validated by both laboratory and field studies. Ting et al. (2021) and Cai et al. (2020) reported that WDA can increase the extent of the convection zone by a factor of two to three compared to submerged zones. Padala et al. (2022) formalized the use of PCC and DPCC as key indices for durability modeling. Furthermore, Cao et al. (2020) demonstrated that the drying-to-wetting ratio is a primary factor governing the depth of the convection zone. Gao et al. (2017) and Liu et al. (2023) developed probabilistic and predictive models that link the depth of the convection zone to environmental wetting-drying cycles.

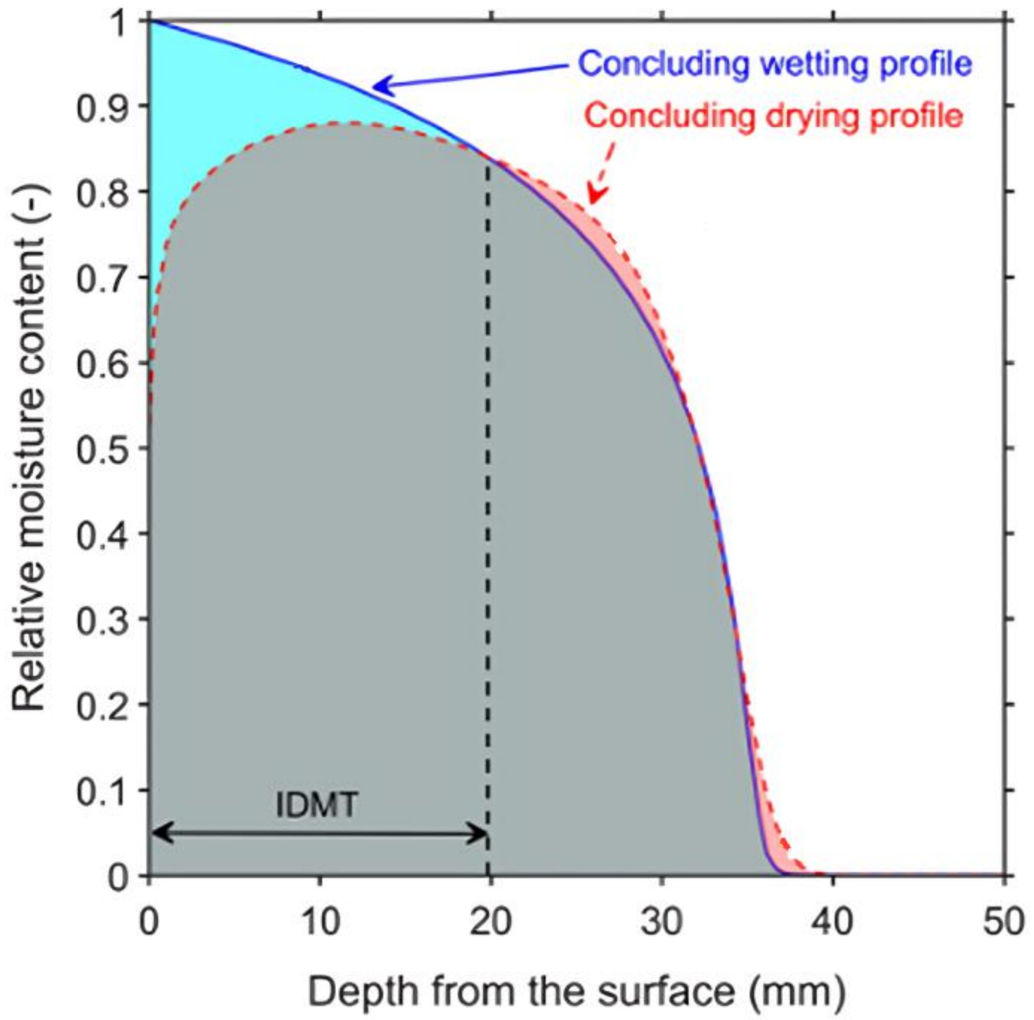

Figure 2.2 Definition of moisture penetration indices from concluding wetting and drying profiles (Adapted from Padala et al., 2022)

2.2 Modeling Approaches

Given the established importance of the convection zone, a significant body of researchers has focused on developing predictive models. These models can be grouped into a hierarchy based on their physical complexity, interpretability, and practical utility. This progression reflects the ongoing effort to balance physical realism with engineering applicability.

2.2.1 Empirical and Analytical Models

The most conventional method of modeling transport in concrete is dependent on empirical models based on Fick's second law of diffusion. The fundamental idea behind this method is that transport is a gradual and slow process powered by an exclusive concentration gradient in a saturated material. These models are very interpretable due to their clear link with physical laws. They also need comparatively small data sets and low computational needs. But their fundamental assumptions prevent them from being applied effectively in wetting-drying (WDA) conditions. As noted by various researchers, such as Li et al. (2022) and Shakouri et al. (2017), these models cannot represent the non-Fickian, convective nature that prevails in the tidal and splash zones. To counter these limitations, researchers created reduced-order and analytical formulations. These frameworks expand the traditional diffusion equation by incorporating an advection term to describe the "pumping" effect of moisture. This strategy, examined by Shi et al. (2023) and Cao et al. (2022), is a better representation of the boundary conditions than the conventional diffusion-only models. Nonetheless, even with this augmentation, they are still simplifications of a very non-linear process. They also fall short in representing the dynamic and time-dependent development of moisture profiles, particularly the variations in Flux and Moisture under fluctuating environmental conditions.

2.2.2 Advanced Numerical Models

The most physically rigorous approach involves advanced numerical models that solve the coupled partial differential equations (PDEs) governing moisture and solute transport. Using techniques such as finite element analysis (FEA), these multi-physics models can account for a wide range of phenomena. These include diffusion, advection, capillary suction, and chemical binding. The work of Padala et al (2022) is particularly noteworthy. They showed that incorporating realistic tidal models for moisture boundary conditions is crucial for accurately predicting key indices such as CZD and CWAA. While these models provide the highest accuracy and retain a moderate degree of interpretability because of their physics-based foundations, their practical use is severely limited. This limitation arises from their very high computational cost and the need for a large number of detailed material and environmental parameters. Such detailed data are rarely available in routine engineering practice, which restricts the adoption of these advanced models.

2.2.3 "Black-Box" Machine Learning Models

The constraints of conventional paradigms have encouraged researchers to investigate data-driven approaches, specifically "black-box" machine learning models like Artificial Neural Networks (ANN) and Support Vector Machines (SVM). Described by Gamil (2023), these models are based on statistical correlation discovery. They apply algorithms to detect intricate, non-linear structures in vast datasets, frequently with high predictive precision. But these models need enormous amounts of data for training and, usually, they have moderate to high computational requirements. Their greatest disadvantage is that they are uninterpretable. The logic within these models is not transparent, i.e., it is not possible to know the justification behind a specific prediction. This transparency makes them hard to rely upon for safety-critical designs and usually results in bad extrapolation when projections are extended to conditions beyond the training set range.

2.2.4 Symbolic Regression

Symbolic Regression (SR) provides a strong complement that wedges together machine learning's ability to predict with traditional physics-based models' interpretability. The fundamental idea behind SR is evolutionary optimization of specific mathematical formulations. As discussed by Makke & Chawla (2024), algorithms like Genetic Programming (GP) navigate the enormous space of potential mathematical formulas to obtain an optimal trade-off between precision and simplicity. The ultimate output of SR is an understandable, human-interpretable equation, which has very high interpretability, a major strength over the black-box methods. While SR can be moderately to heavily data- and computationally intensive, it completely sidesteps the interpretability issue. It comes with its own set of issues, though. The evolutionary search can be highly computationally expensive, and there is always a potential of returning extremely complex or even physically vacuous solutions that are not particularly insightful.

The evolution of such modeling paradigms is represented in **Figure 2.3**, and their essential features are tabulated in **Table 1**. These two figures collectively present in brief the compromise between interpretability, data needs, and computational expense that directed the evolution of durability modeling in marine-exposed concrete.

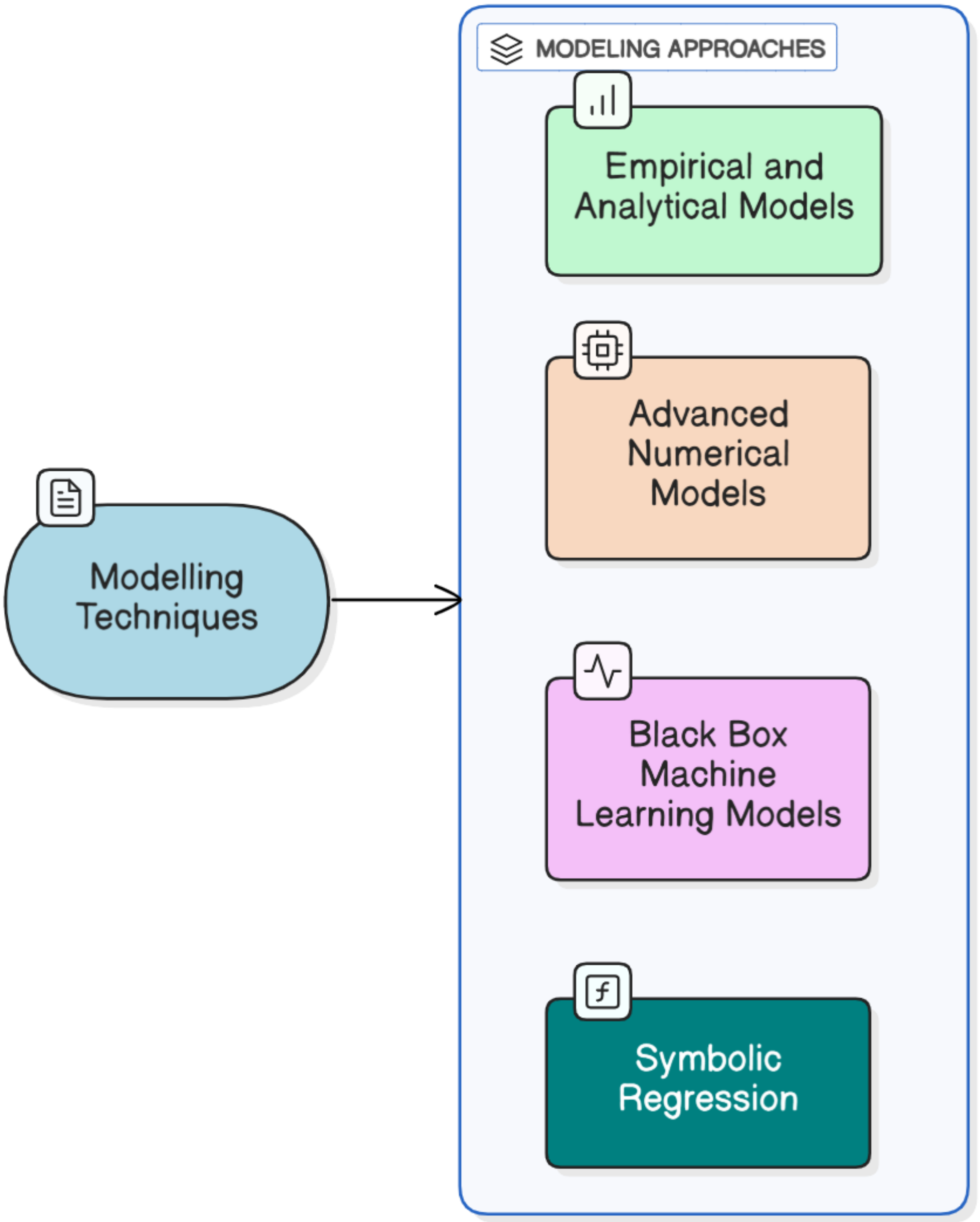

Figure 2.3 The types of modeling paradigms for transport phenomena in concrete

Table 1 A Comparative Summary of Modeling Paradigms

Modeling Paradigm	Underlying Principle	Interpretability	Data Requirement	Computational Cost	Key Limitations
Empirical analytical	Fick's Second Law of Diffusion.	An explicit equation with fitted parameters. Highly interpretable.	Small datasets sufficient for curve fitting a few parameters.	Minimal; often solvable with simple calculations.	Fails to capture non-Fickian behavior; assumes saturation.
Advanced numerical	Multi-physics transport equations.	A numerical simulation result; Moderately interpretable but complex.	Extensive set of material properties; detailed environmental BC's.	Require specialized software; significant runtime.	difficult to calibrate for field conditions.
Black-box machine learning	Statistical correlation finding.	An opaque 'black-box' model; Very low interpretability.	Very large and diverse datasets required.	Moderate to high for training; fast for prediction.	difficult to trust for safety-critical design; poor extrapolation.
Symbolic Regression	Evolutionary search for mathematical expressions.	An explicit, human-readable mathematical equation. Very high interpretability.	Moderate to large datasets; benefits from well-structured data.	evolutionary search can be intensive.	susceptible to finding overly complex solutions.

2.3 Research Gaps

The discussion above points to a key shortcoming in marine concrete structure durability modeling. The essential function of moisture transport in creating the convection zone is adequately established. Indices like CZD has been found to describe this zone. Yet, there still lacks a workable and accurate predictive model for the convection zone depth (CZD). Current models offer an explicit trade-off:

- Empirical and analytical models are easy to apply and straightforward. They are not, however, as accurate or generalizable as needed for sound design in variable wetting-drying (WDA) situations.
- Complex numerical (PDE) models, on the other hand, are precise and more fully represent the underlying physics. However, they are computationally costly and demand many fine-scale input parameters. These data are not available in standard engineering practice, and such models are not applicable for any common real-world situation.
- Black-box" machine learning models are capable of good predictive performance. However, they are not interpretable and so are inappropriate for use in safety-critical applications where engineers need to comprehend and rely on the reasoning for the predictions.

In addition, while some research has qualitatively correlated CZD with wetting (t_{wet}) and drying (t_{dry}) cycle durations, no explicit, proven mathematical relationship defining such a correlation exists that also accounts for the environmental severity parameters of Flux ($\mathrm{q}_{\mathrm{dry}}$) and Moisture ($\theta_{r,\,eq}$). The absence of such a correlation hinders engineers from introducing the particular site-specific cyclic exposure conditions into service-life estimations in a logical manner.

2.4 Motivation and Objectives

This research is motivated by the need to develop an interpretable and physically meaningful analytical framework for predicting convection zone evolution in marine concrete structures subjected to cyclic wetting and drying exposure conditions. Existing approaches often rely either on computationally expensive numerical simulations or empirical fitting methods that provide limited physical interpretability.

The present study aims to bridge this gap by combining Symbolic Regression, reduced-order coefficient analysis, DWTR-based transport characterization, and Machine Learning validation to identify the dominant physical mechanisms governing moisture transport behavior.

Rather than focusing solely on predictive accuracy, the research emphasizes the discovery of generalized transport relationships capable of representing the transition between drying-controlled and wetting-controlled exposure regimes under varying marine environmental conditions.

The primary objectives of the present study are summarized as follows:

- To apply Symbolic Regression (PySR) for identifying physically interpretable mathematical relationships governing Convection Zone Depth (CZD) evolution under cyclic marine exposure conditions.
- To investigate the relative influence of the governing environmental variables, including wetting duration (t_{wet}), drying duration (t_{dry}), drying flux (q_{dry}), and equilibrium moisture ($\theta_{r,eq}$), on convection-zone transport behavior.
- To develop a reduced-order normalized transport framework based on dominant temporal exposure variables and extract physically meaningful transport coefficients representing drying-controlled and wetting-controlled mechanisms.
- To establish generalized DWTR-based coefficient relationships capable of describing the nonlinear transition between drying-dominated and wetting-dominated transport regimes across varying marine exposure conditions.
- To independently validate the identified transport mechanisms using Random Forest-based feature importance analysis and evaluate the robustness of the proposed reduced-order modelling framework.
- To provide a simplified yet physically interpretable analytical framework capable of bridging the gap between computationally intensive numerical simulations and highly simplified empirical transport models used in marine durability analysis.

Chapter 3
Methodology

In this chapter, the method utilized to develop an empirically based predictive model for CZD under cyclical marine exposure is discussed. In this regard, A Symbolic Regression (SR) framework implemented through PySR considering employing four parameters: wetting time (t_{wet}), drying time (t_{dry}), drying rate ($\mathrm{q}_{\mathrm{dry}}$), and equilibrium moisture content ($\theta_{r,\,eq}$). The outcomes showed that the selected parameters had significant impacts with variations in different stations and exposure cycles. Consequently, the concept of parameter isolation, threshold minimization, and equation simplification was developed to achieve simplicity in interpretation and robustness in modeling. Finally, symbolic regression and curve fitting techniques were employed in this work to derive predictive equations for each level and station.

3.1 Dataset Description

The foundation for this study is a structured dataset. It was designed to analyze the relationship between cyclic environmental exposure and the resulting Convection Zone Depth (CZD). The data is organized to simulate varied conditions that represent different geographical locations. As shown in **Figure 3.1**, the dataset is sourced from 36 distinct coastal ports across India, some of which are shown in Figure 3.1.

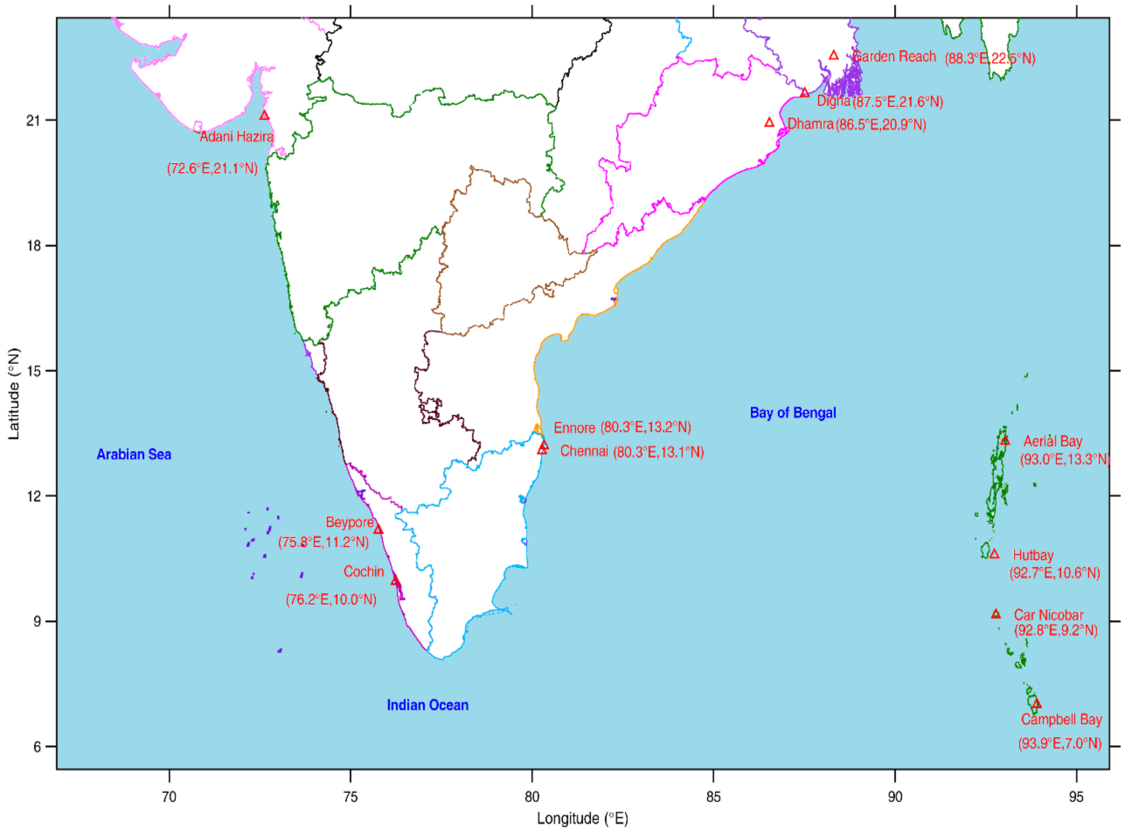

Figure 3.1 Ports Marked in Indian Map

For each port, the dataset contains information corresponding to 10 different exposure levels, which represent varying sea levels or specific tidal conditions. This structure results in 360 individual data files. The initial step of the methodology involved programmatically loading and consolidating these files into a single, unified data frame. This master dataset served as the foundation for all subsequent modeling tasks.

Each data point in the consolidated dataset contains 5 columns. These consist of four independent variables (input features) and one dependent variable (the target for prediction):

- **Input Feature 1 (t_{wet}):** The first input feature, t_{wet}, represents the duration of the wetting period in each cycle (in hours). It is a critical parameter that governs the ingress of moisture into the concrete through capillary action.
- **Input Feature 2 (t_{dry}):** The second input feature, t_{dry}, quantifies the duration of the drying period in each cycle (in hours). During this phase, moisture evaporates from the near-surface region, which leads to the accumulation of any dissolved substances within the concrete's pore structure.

- **Input Feature 3 (q_{dry}):** The Drying Flux, representing the rate of ion ingress from the environment (taken constant i.e. median).
- **Input Feature 4 ($\theta_{r,\,eq}$):** The Equilibrium Moisture, representing the internal saturation state of the concrete (taken constant i.e. median).
- **Target Variable (CZD):** The target variable, CZD, represents the Convection Zone Depth (in mm). It is the measured depth from the exposed concrete surface that is influenced by cyclic moisture movement.

3.2 Feature Engineering

The CZD formation mechanism involves complicated interplay among the wetting-drying cycle exposure. Since the transport response is nonlinear in nature, the Physics-Informed Feature Engineering method was used in order to help the model recognize important associations between the environmental factors and the CZD.

The environmental data was readily accessible at hourly intervals; however, the wetting-drying cycle exposure was recorded event-wise and had varying lengths. As such, a weighted time window averaging method was employed to align the environmental data with each individual exposure cycle. Furthermore, the cyclic time loop technique was used in order to simulate the structural exposure over the course of a year based on annual environmental data.

Initially, the Symbolic Regression framework was trained using four environmental variables: wetting duration (t_{wet}), drying duration $\left(t_{dry}\right)$, drying flux $\left(q_{dry}\right)$, and equilibrium moisture $\left(\theta_{r,eq}\right)$. However, preliminary analysis revealed stronger dependence on the temporal variables $\left(t_{wet}, t_{dry}\right)$. Therefore, during later stages of the study, $q_{dry}$and $\theta_{r,eq}$ were constrained around their median values to reduce environmental variability and enable the development of simplified equations.

Moreover, the Symbolic Regression technique created additional mathematical functions by applying various mathematical operators and optimized constants to the input features.

3.3 Dataset Preprocessing and Analysis

Before the dataset could be used for model training, a series of preprocessing steps were essential. These steps cleaned, normalized, and structured the data appropriately. This phase ensures that the model receives high-quality, consistent input, which is critical for achieving reliable and accurate performance.

3.3.1 Data Cleaning

The provided dataset was of high quality and did not contain missing or anomalous values. Therefore, no data imputation or removal was required. The primary cleaning step involved verifying the data types and ensuring consistency across the consolidated dataset.

3.3.2 Feature Scaling

Features with larger numerical ranges can disproportionately influence a model's learning process. While several scaling approaches exist, Min–Max Scaling is particularly useful when the goal is to map all features into a consistent, bounded range. To address the varying scales of the input variables in the dataset, Min–Max Scaling was applied.

The Min–Max Scaler rescales each feature to a fixed interval, typically [0, 1], using the formula:

$$\boldsymbol{X_i}' = \frac{\boldsymbol{X_i} - \boldsymbol{min}(\boldsymbol{X})}{\boldsymbol{max}(\boldsymbol{X}) - \boldsymbol{min}(\boldsymbol{X})} \qquad \text{Eq. (3.3.2)}$$

Where:

- X_{min}= minimum value of the feature
- X_{max}= maximum value of the feature

This transformation preserves the shape of the original distribution while ensuring that no feature dominates due to magnitude differences. It also makes the model more stable during optimization and improves convergence when features operate on similar scales.

3.3.3 Data Splitting

To ensure robust model training and an unbiased final evaluation, the consolidated dataset was partitioned into two distinct subsets. This is a standard and critical practice in machine learning to prevent model overfitting.

- Training Set (80%): The training set, comprising 80% of the total data, was used to train the models. This allowed the algorithm to learn the underlying patterns and relationships between the input features and the target variable, CZD.

- Test Set (20%): The test set, consisting of the remaining 20% of the data, was held out during the training process. It was used only once, after all model development was complete, to provide an objective and unbiased assessment of the final model's performance on unseen data.

The split was performed in a stratified manner. This ensures that the statistical distribution of the target variable and key features was preserved across both subsets, making the evaluation process more reliable.

3.3.4 Targeted Parameter Isolation

Initial symbolic regression analyses indicated tests utilizing all four variables indicated that the time-related parameters (t_{dry}, t_{wet}) had relatively higher impact on the dynamics of the CZD compared to the other two parameters, q_{dry} and $\theta_{r,eq}$ Also, results from the FEM simulation indicated distinct stepwise changes in the depths due to numerical modeling in the mesh.

To facilitate isolation of the physical relationships and improve interpretability of equations generated, a selective approach to isolating the governing parameters was adopted.

- **Isolation of Controlled Variables**: Environmental parameters such as q_{dry} and $\theta_{r,eq}$ were kept within the median range to ensure minimal interference of the other environmental factors. Consequently, the analysis focused primarily on the interplay of the wetting and drying time periods.
- **Drying-to-Wetting Time Ratio (DWTR):** To compare the environmental exposure behavior across different coastal stations and exposure levels, the Drying-to-Wetting Time Ratio (DWTR) was introduced as an additional comparative parameter. DWTR was defined as:

$$DWTR = \frac{t_{dry}}{t_{wet}}$$

- **Analysis by Level and Location**: Following the isolation of the variables, each dataset underwent an independent analysis on a station-to-station and level-wise basis.

This process greatly improved the interpretability of generated symbolic expressions and facilitated the development of simplified curve fitting models.

3.4 Model Development

In this work, Symbolic Regression (SR) has been used as the main method of analysis due to the ability of this machine learning algorithm to find mathematical dependencies between different quantities directly from data sets. The key advantage of symbolic regression over other regression algorithms lies in its capability to find the form of the equation along with its coefficients at once.

3.4.1 The Symbolic Regression Paradigm

The SR process is guided by a Genetic Programming (GP) algorithm. GP is an evolutionary method which implements natural selection to "evolve" a population of potential mathematical expressions. The algorithm starts with an initial population of randomly generated equations. In every generation, the algorithm chooses the top-performing equations according to a fitness measure. It then crosses them over via a "crossover" operation and adds random alterations using "mutation" to generate a new generation of offspring equations. This process iterates so that the population converges towards solutions that provide a more optimal balance between precision and simplicity.

3.4.2 The PySR Framework

For this research, SR was applied through PySR, a cutting-edge, open-source Python package. PySR has a very optimized Julia backend for speeding up the evolutionary search. This makes it possible to sample an enormous and intricate search space. One of the benefits of PySR is that it places particular emphasis on generating parsimonious models. These are formulas that are as uncomplicated as can be achieved without losing important accuracy. This is vital when engineering solutions that require interpretability to be on par with prediction capability.

For this study, specific configurations were applied to handle the physical constraints of the problem:

- Safe Operators: To prevent mathematical instability (e.g., taking the root of a negative number created by scaling), custom "safe" operators were defined: sqrt_abs [$\sqrt{|x|}$] and log1p_abs [$\log(1 + |x|)$].
- Robust Loss Function: The fitness of equations was evaluated using Mean Absolute Error (L1 Loss) instead of Mean Squared Error (L2 Loss). This

reduced the impact of outliers, ensuring the discovered equations represent the general physical behavior of the port.

3.4.4 The PySR Equation Discovery Workflow

The evolutionary approach taken by PySR involves discovering mathematical equations directly from data. The workflow that has been chosen for this paper is presented schematically in Figure 3.3 and entails the following stages:

- Initialization – a random selection of mathematical expressions is created using the input variables and mathematical functions.
- Fitness Evaluation – fitness values of all mathematical equations are calculated according to prediction performance on the training dataset and their complexity.
- Selection – the most fitting equations are chosen to participate in further generations of the search.
- Genetic Operations – novel mathematical expressions are generated using crossover and mutation operations.
- Optimization of Constants – mathematical expressions are optimized using numerical methods to determine the optimal constants.
- Repetition – genetic algorithm is repeated several times until the desired result is achieved.
- Output – a set of mathematical equations is returned.

Figure 3.3 shows a graphical representation of the workflow.

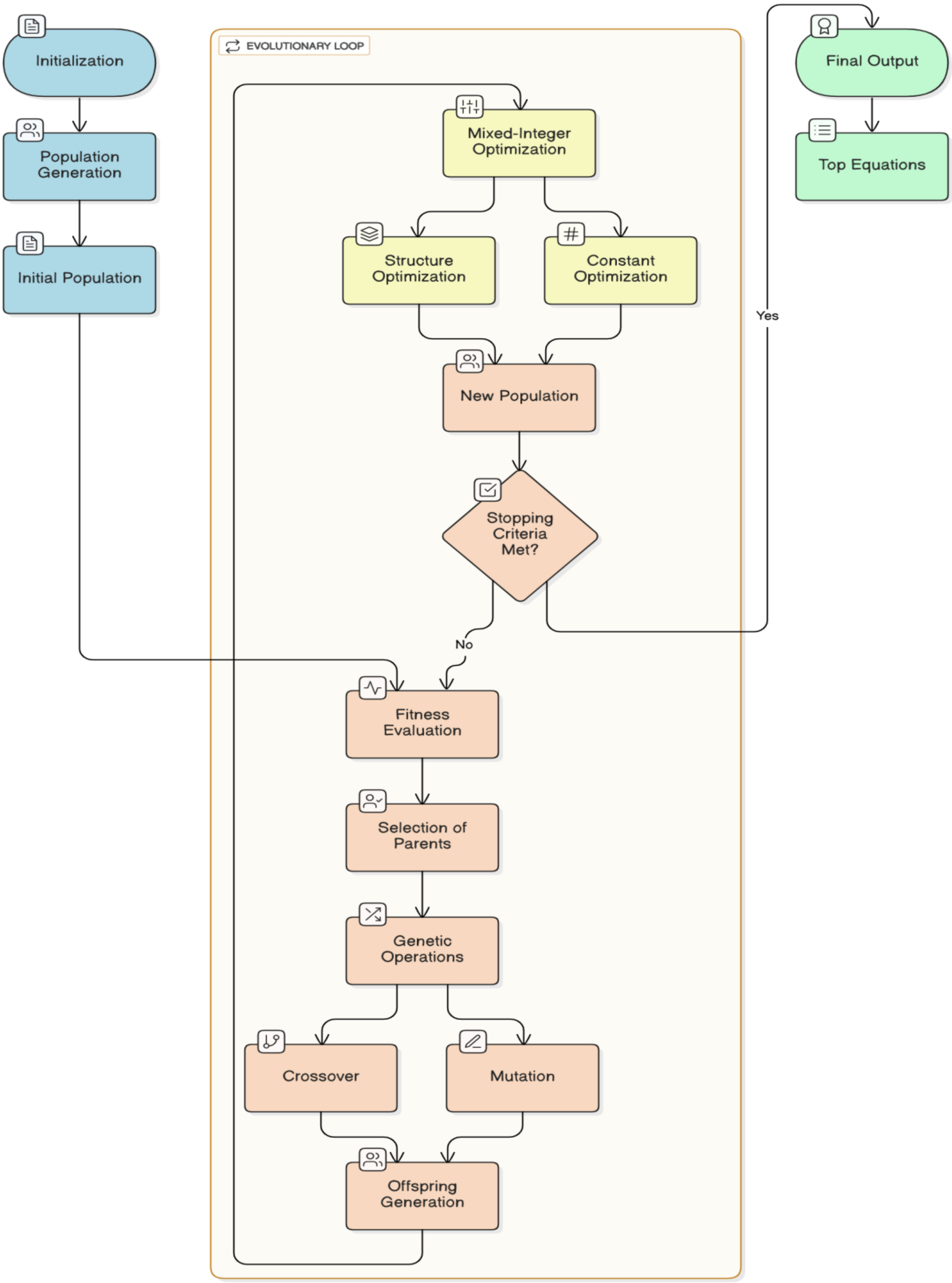

Figure 3.2 Workflow of PySR

3.4.5 Search Space and Hyperparameters

The PySR search process was configured to identify physically meaningful and interpretable equations for CZD prediction.

- Search Space: The algorithm was given the entire set of fundamental and engineered features as constructions. The search space was also established by a rich collection of mathematical operators, such as simple arithmetic (+, *), and non-linear ones (log, sqrt).
- Fitness Function: Equation fitness was evaluated using Mean Absolute Error (L1 Loss), while complexity penalties were applied to encourage simpler and more interpretable symbolic expressions.
- Hyperparameter Configuration: The evolutionary search process was controlled using parameters such as population size, number of iterations, operator probabilities, and equation complexity limits.

This configuration enabled the framework to balance predictive performance with equation simplicity during symbolic equation discovery.

3.4.6 Simplified Equation Development

While the first stage involved modeling based on all four environmental variables, subsequent research into symbolic regression involved creating simpler and more meaningful mathematical relationships. The second step was to set the variables q_{dry} and $\theta_{r,eq}$ at their medians due to the presence of secondary variation, with an emphasis on exposure time parameters.

From the symbolic trends identified during the PySR search process, simplified low-complexity equations were developed using the variables t_{wet} and t_{dry} Square-root relationships in mathematical modeling were used to incorporate cycles without losing simplicity and numeracy.

These simplified equations were meant to act as a reduced predictive model that could be assessed individually for various levels of exposure.

3.4.7 Curve Fitting and Coefficient Optimization

Having extracted simplified symbolic equation forms using the PySR framework, regression-based curve fitting methods were performed to optimize the estimated coefficients associated with the equations. Simplified equations concentrated on the two main time-related variables, namely t_{wet} and t_{dry}, with q_{dry} and $\theta_{r,eq}$ fixed near their medians during the isolation phase.

The process of optimizing the coefficients included regression-based fitting of simplified equations to the actual observed CZD using regression methods that minimize prediction errors for response variables. Linear and nonlinear regression-based fitting methods were performed based on the structure of the extracted symbolic equations.

Specifically, simplified equations of the following forms:

$$\boldsymbol{CZD} = \boldsymbol{k_1}\sqrt{\boldsymbol{T_{dry}}} + \boldsymbol{k_2}\sqrt{\boldsymbol{T_{wet}}} \qquad \text{Eq. 3.4.1}$$

were fit to the data for some selected datasets at station-level and level-wise to obtain the optimal values of coefficients (k1, k2).

3.5 Model Evaluation

To measure the ability of the models generated from the symbolic regression in predicting future outcomes, a quantitative evaluation framework was built. The found symbolic equations and simplifications to curve fits were tested on out-of-sample data to get unbiased results from various environmental conditions.

3.5.1 Validation strategy

Both the outputs from symbolic regression analysis and machine learning-based validation were considered while testing the dataset. Apart from the simplified mathematical equations found through PySR and curve fits, a Random Forest Regressor was also used as an auxiliary validation method for capturing complex relationships in the dataset.

The methods utilized to validate the dataset are mentioned below:

- **Scaling of Variables**: To make sure that the numbers stay within certain limits when the model is running, all variables were scaled using Min–Max Scaling.
- **Random Forest Regression**: For understanding any complex nonlinear relationship in the dataset, a Random Forest Regressor was implemented.

- **Importance of Environmental Variables**: The Random Forest model was further used to find the relative importance of the environmental variables affecting CZD.

The above-mentioned validation methods made it possible to compare the output of both types of models.

3.5.2 Metrics

A primary metric was used for this evaluation: the Coefficient of Determination (R2).

- **Coefficient of Determination (R2):** This score measures the proportion of the variance in the target variable (CZD) that is predictable from the input features. An R2 value closer to 1.0 indicates a better fit. It is calculated as:

- **Root Mean Square Error:** RMSE measures the average magnitude of

$$\boldsymbol{R2 = 1 - \frac{\sum_{i=1}^{n}(y_i - \hat{y}_i)^2}{\sum_{i=1}^{n}(y_i - \bar{y})^2}} \qquad \text{Eq. 3.5.1}$$

prediction error between observed and predicted CZD values**.**

$$\mathbf{RMSE} = \sqrt{\frac{\sum_{i=1}^{n}(y_i - \hat{y}_i)^2}{n}} \qquad \text{Eq. 3.5.2}$$

Where y_i is the actual observed CZD for the ith data point, $\hat{y}_i$ is the predicted CZD from the model, $\bar{y}$ is the mean of the observed CZD values, and n is the number of data points in the test set. This metric provides a comprehensive benchmark for assessing the accuracy and reliability of the final predictive equation

Chapter 4
Results and Discussion

This chapter discusses the outcomes of the multi-phase analytical modeling process. Initially, Symbolic Regression (PySR) was deployed to discover baseline equations linking environmental exposure to the Convection Zone Depth (CZD). However, the discovery of systematic variance anomalies and discretization artifacts inherent to the FEM mesh motivated refinement of the analytical framework. This chapter details the progression from baseline anomaly detection to establishing parameter dominance using a Random Forest model, culminating in the extraction of station-level physical coefficients and the identification of universal moisture transport trends via the Drying-to-Wetting Time Ratio.

4.1 Initial SR Exploration

The initial stage of the analysis involved applying the PySR Symbolic Regression framework independently across different exposure levels to identify mathematical relationships governing the Convection Zone Depth (CZD). During this exploratory phase, all environmental variables were supplied to the symbolic search process without imposing any predefined equation structure. his stage aimed to investigate whether physically meaningful relationships could emerge directly from the dataset and to examine how these relationships varied across different exposure conditions.

PySR generated a series of symbolic equations for individual exposure levels of the selected coastal stations. The baseline equations obtained for AdaniHazira are summarized in Table 2.

Table 2 Equation discovered for AdaniHazira

Level	Equation	R2 score
1	$5.46\sqrt{t_{dry}}$	0.65
2	$7.09\sqrt{t_{dry}}$	0.92
3	$8.43\sqrt{t_{dry}}$	0.90
4	$11.66\sqrt{t_{dry}}$	0.89
5	$13.94\sqrt{t_{dry}}$	0.79
6	$21.27\sqrt{t_{dry}}$	0.55
7	$26.45\sqrt{t_{dry}} + 4.98\, t_{wet}$	0.78
8	$0.027\sqrt{t_{dry}t_{wet}}$	0.50
9	$26.67\sqrt{t_{wet}}$	0.52
10	$24.16\sqrt{t_{wet}}$	0.60

The generated equations exhibited varying mathematical forms and predictive accuracies across different levels. Several levels produced compact symbolic expressions with comparatively high R^2scores, indicating that meaningful physical relationships were successfully identified by the symbolic analysis.

4.1.1 Identification of Dominant Temporal Variables

Analysis of the generated equations revealed that the temporal exposure variables played the dominant role in predicting CZD behavior. In particular, the drying duration parameter (t_{dry}) repeatedly appeared in the discovered symbolic forms, especially through square-root relationships.

For upper and intermediate exposure levels, the equations were primarily governed by terms involving $\sqrt{t_{dry}}$, indicating strong dependence on drying duration under cyclic exposure conditions. In contrast, the lower exposure levels located closer to the submerged region increasingly incorporated wetting duration (t_{wet})within the symbolic expressions. The repeated occurrence of these temporal parameters across multiple independently generated equations indicated that cyclic wetting-drying behavior strongly influenced the convection zone formation process.

The predictive accuracy also varied across different levels, suggesting localized environmental variability and level-dependent transport behavior within the dataset. Localized station-level analyses were additionally performed to further investigate recurring symbolic structures and improve equation interpretability

4.1.2 Interpretation of Recurring Symbolic Patterns

Beyond predictive performance, the symbolic regression process revealed recurring mathematical structures across multiple levels. Square-root relationships involving t_{dry} appeared consistently in several equations, suggesting a nonlinear dependence between drying duration and CZD evolution.

The emergence of similar symbolic patterns across independently trained level-wise models indicated that the framework was identifying stable physical trends rather than unstable mathematical artifacts. These recurring structures motivated the formulation of simplified reduced-order equations based on dominant temporal variables.

The observations obtained from this exploratory symbolic regression stage formed the foundation for the normalized equation development and coefficient-based analyses discussed in the subsequent sections.

Based on the recurring symbolic patterns identified across multiple exposure levels, a simplified normalized equation framework was formulated to represent the dominant cyclic exposure behavior governing CZD formation. The proposed reduced-order relationship was expressed as given in equation no. 3.4.1.

where $k_1$and $k_2$represent the coefficients associated with drying-controlled and wetting-controlled transport behavior, respectively. This formulation provided a compact and interpretable representation of the dominant temporal effects identified through symbolic regression while significantly reducing equation complexity.

4.2 Curve Fitting & Coefficient Extraction

Following the formulation of the normalized reduced-order equation, regression-based curve fitting procedures were performed to estimate the coefficients associated with the dominant temporal variables. The objective of this stage was to transform the symbolic patterns identified through PySR into compact predictive relationships that could be evaluated consistently across different stations and exposure levels.

The normalized equation framework adopted for the fitting process was expressed as equation 3.4.1.

The fitting process was performed using regression-based optimization techniques to minimize prediction error between observed and predicted CZD values.

4.2.1 Regression-Based Coefficient Optimization

Regression-based curve fitting was performed independently across selected datasets to estimate the optimal values of k_1 and k_2. The fitting procedure utilized the simplified normalized equation obtained from the symbolic regression analysis and optimized the coefficients based on the observed CZD response.

The simplified formulation reduced equation complexity while preserving the dominant transport behavior identified during the exploratory symbolic analysis. The resulting fitted equations provided compact and interpretable representations of CZD evolution under cyclic marine exposure conditions.

Although the initial symbolic regression stage included additional environmental variables such as drying flux $\left(q_{dry}\right)$and equilibrium moisture parameter $\left(\theta_{r,eq}\right)$, subsequent analyses indicated that their influence remained comparatively smaller than the temporal exposure variables $\left(t_{dry}, t_{wet}\right)$. To obtain a generalized reduced-order formulation suitable for cross-station comparison, representative constant values for these secondary parameters were adopted using median environmental conditions across the analyzed datasets. This simplification enabled the final analytical framework to focus primarily on the dominant cyclic wetting-drying transport behavior while maintaining model stability and interpretability.

A representative comparison between observed and fitted CZD values for a particular station's level is shown in Figure 4.1.

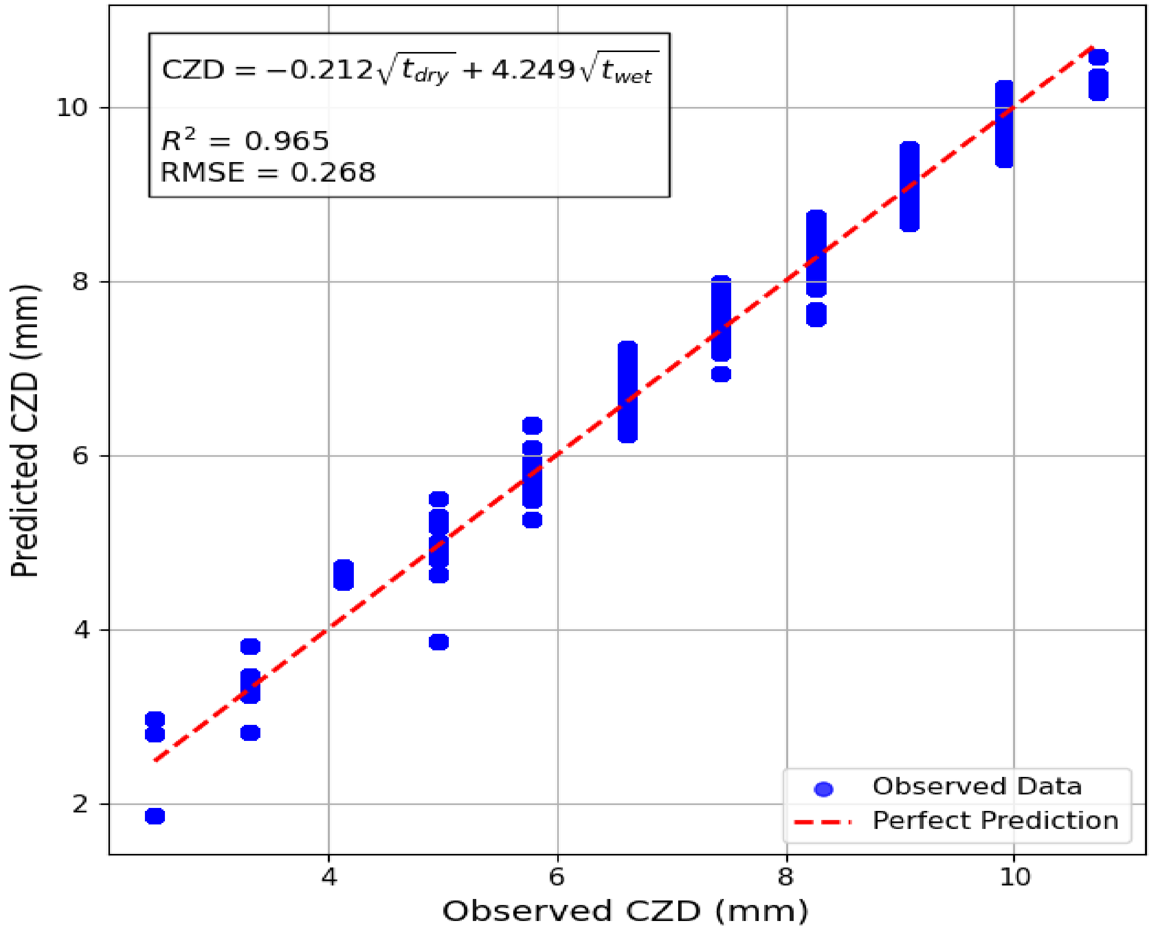

Figure 4.1 Comparison Between Observed and Predicted CZD Values Using the Reduced-Order Regression Model

4.2.2 Station-Wise Coefficient Estimation

The regression fitting process was applied independently across different coastal stations to obtain localized coefficient estimates. This station-wise fitting approach enabled the normalized equation to adapt to variations in environmental exposure conditions across different geographic locations.

The extracted coefficient values demonstrated noticeable variation between stations, indicating differences in the relative influence of drying and wetting behavior under varying marine exposure environments.

The estimated station-wise coefficients are summarized in Table 3.

Table 3 Coefficient Value for AdaniHazira

Station	Level	k1	k2	R2	RMSE
S1	1	3.20	-0.01	0.59	0.82
S1	2	3.72	-0.02	0.95	0.29
S1	3	3.94	-0.07	0.95	0.28
S1	4	4.16	-0.15	0.95	0.26
S1	5	4.29	-0.20	0.91	0.27
S1	6	3.74	0.44	0.75	0.40
S1	7	3.62	0.91	0.81	0.99
S1	8	2.03	4.67	0.45	3.17
S1	9	0.38	11.58	0.55	4.07
S1	10	0.18	14.45	0.85	1.46

[The complete set of extracted level-wise and station-wise regression coefficients used in the DWTR-based analysis is provided in the Appendix for reference.]

4.2.3 Level-wise Coefficient Estimation

In addition to station-wise fitting, coefficient estimation was also performed independently across different exposure levels. This level-wise analysis enabled investigation of how the relative contributions of drying- and wetting-controlled transport mechanisms varied throughout the exposure profile.

The extracted coefficients revealed systematic transitions between drying-dominated and wetting-dominated behavior across different levels, consistent with the symbolic patterns observed during the initial PySR exploration. These fitted coefficient datasets formed the basis for the subsequent DWTR-based trend analysis discussed in the following section.

4.3 DWTR - Based Coefficient Analysis

Following the extraction of the level-wise regression coefficients, an additional analysis was performed to investigate the relationship between the fitted temporal coefficients and the cyclic exposure characteristics of the marine environment. For this purpose, the Drying-to-Wetting Time Ratio (DWTR) was introduced as a governing exposure descriptor.

The DWTR-based analysis enabled comparison of the relative dominance of drying- and wetting-controlled transport behavior across different stations and exposure levels. By relating the extracted coefficients to the cyclic exposure ratio, broader physical trends governing CZD evolution could be identified beyond individual level-wise curve fitting results.

4.3.1 Definition of DWTR

The Drying-to-Wetting Time Ratio (DWTR) was defined as:

$$DWTR = \frac{t_{dry}}{t_{wet}}$$

where:

- t_{dry} represents the drying duration of the cyclic exposure period,
- t_{wet} represents the wetting duration.

This parameter was introduced to characterize the relative dominance of drying and wetting phases within each exposure cycle. Larger DWTR values correspond to drying-dominated exposure conditions, while lower DWTR values indicate increased wetting influence.

For each dataset, the representative DWTR value was computed using the average temporal durations associated with the corresponding station and exposure level.

4.3.2 Variation of coefficients with DWTR

The extracted transport coefficients exhibited systematic nonlinear variation with the Drying-to-Wetting Time Ratio (DWTR). To investigate the consistency of this behavior across different coastal environments, the fitted coefficients obtained from all analysed stations were plotted against the corresponding DWTR values.

Figure 4.2 presents the variation of the drying-related coefficient k_1 and the wetting-related coefficient k_2 for multiple stations together with the final generalized coefficient relationships represented by the black dashed curves.

The drying-related coefficient k_1 exhibited comparatively large magnitudes at lower DWTR values and progressively decreased with increasing DWTR. The transition was particularly sharp within the intermediate DWTR range, after which the coefficient gradually approached near-zero values under highly drying-dominated exposure conditions. This behavior indicates that the contribution of drying-controlled transport mechanisms reduces significantly as the cyclic exposure ratio increases.

In contrast, the wetting-related coefficient k_2 demonstrated a distinctly different nonlinear trend. At very low DWTR values, the wetting-related contribution remained relatively small. However, as DWTR increased, k_2 rapidly increased and reached a broad peak within the intermediate cyclic exposure regime. Beyond this transitional region, the coefficient gradually reduced again at larger DWTR values.

The black dashed fitted curves closely followed the overall station-wise transition behavior, despite localized variability between individual datasets. This consistency suggested the presence of generalized transport trends governing convection zone evolution across different marine exposure environments.

The observed transition behavior indicates that convection-zone transport mechanisms do not vary linearly with exposure conditions, but instead shift progressively between drying-controlled and wetting-controlled regimes depending on the cyclic exposure ratio.

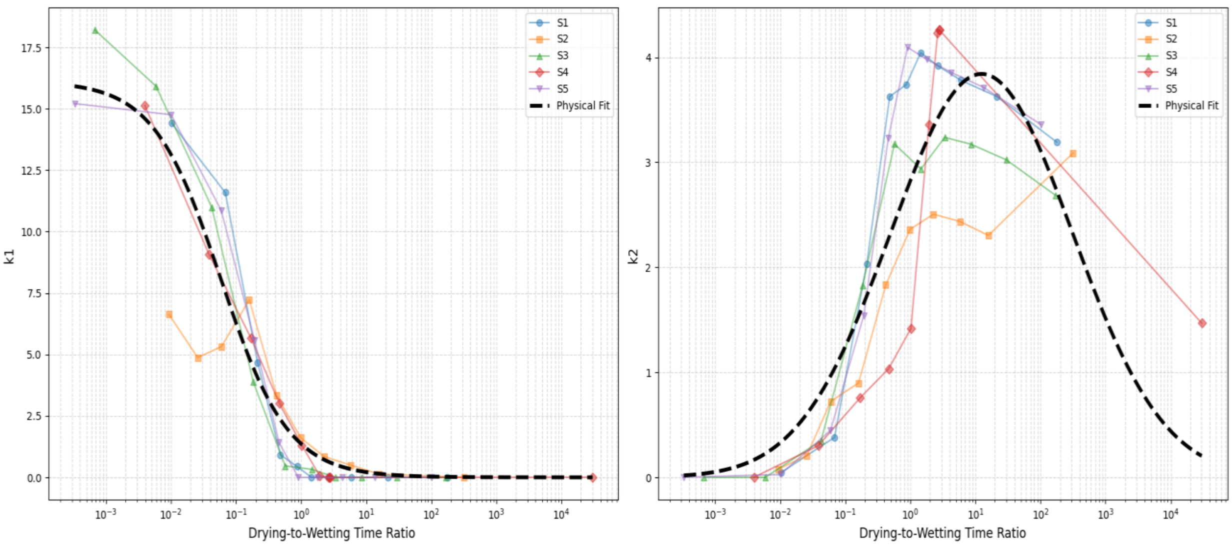

Figure 4.2 Variation of Drying-Controlled (k_1) and Wetting-Controlled (k_2) Transport Coefficients with DWTR Across Different Coastal Stations

4.3.3 Comparative Analysis Across Stations and Levels

Comparison of the extracted coefficients across different stations and exposure levels revealed systematic transitions in the governing transport behavior throughout the exposure profile.

Upper exposure regions generally exhibited stronger drying-controlled characteristics, reflected through larger k_1 contributions and relatively smaller wetting-related coefficients. Conversely, lower exposure regions located closer to submerged conditions demonstrated increasing wetting dominance through progressively larger k_2 values.

Although the exact coefficient magnitudes varied across stations due to localized environmental variability, the overall transition trends remained consistent across all analysed datasets. The clustering of the station-wise datasets around the generalized fitted curves demonstrated that the normalized reduced-order equation framework successfully captured the dominant cyclic transport mechanisms governing CZD evolution under marine exposure conditions.

The transition behavior additionally aligned with the symbolic patterns identified during the initial PySR exploration stage. Exposure levels dominated by drying behavior repeatedly produced symbolic equations involving $\sqrt{t_{dry}}$, whereas lower exposure levels increasingly incorporated wetting-related terms involving t_{wet}.

This DWTR-based coefficient analysis therefore established a direct physical linkage between the symbolic regression findings, normalized coefficient extraction, and the governing cyclic exposure characteristics of the marine environment.

4.3.5 Universal Coefficient Function Development

To generalize the transport behavior observed across different stations and exposure levels, nonlinear regression fitting procedures were applied to the extracted DWTR-based coefficient datasets. The objective of this stage was to formulate continuous analytical relationships capable of representing the evolution of the drying- and wetting-related transport coefficients under varying cyclic exposure conditions.

The generalized fitted relationships are represented by the black dashed curves shown in Figure 4.2.

The drying-related coefficient $k_1$demonstrated a monotonic nonlinear decay trend with increasing DWTR. To capture this transition behavior analytically, a sigmoidal-type nonlinear fitting function was adopted:

$$k_1(DWTR) = \frac{16.63}{1 + 19.4(DWTR)^{1.27}}$$

The fitted relationship showed strong agreement with the extracted coefficient dataset and successfully reproduced the rapid transition between drying-dominated and wetting-influenced cyclic exposure regimes.

Similarly, the wetting-related coefficient $k_2$exhibited a nonlinear peak-type response across the DWTR range. A Gaussian-type nonlinear fitting function was therefore adopted:

$$k_2(DWTR) = 4.07e^{\frac{(\ln DWTR - 2.37)^2}{2(3.2)^2}}$$

The fitted $k_2$relationship successfully captured the broad transitional peak observed in the station-wise datasets, indicating the existence of an intermediate cyclic exposure regime where the interaction between wetting and drying durations maximally influences convection-zone evolution.

The resulting analytical relationships provide a generalized framework for estimating the dominant cyclic transport coefficients directly from exposure-cycle characteristics. These fitted coefficient laws further demonstrate that convection-zone transport behavior transitions progressively between drying-controlled and wetting-controlled regimes depending on the cyclic exposure ratio.

4.4 Independent Validation of Dominant Transport Mechanisms

To independently validate the dominant transport mechanisms identified through symbolic regression and DWTR-based coefficient analysis, an additional Machine Learning validation framework was implemented using a Random Forest Regressor. The objective of this stage was not to derive explicit predictive equations, but rather to quantitatively evaluate the relative influence of the environmental exposure variables governing CZD evolution.

The Random Forest model was trained using the complete environmental dataset containing the four governing exposure variables:

$$t_{dry},\ t_{wet},\ q_{dry},\ \theta_{r,eq}$$

Prior to model training, all variables were normalized using Min–Max scaling to ensure consistent numerical treatment during the learning process.

4.4.1 Feature Importance Analysis

Feature importance values extracted from the Random Forest model provided quantitative insight into the relative contribution of each environmental parameter toward CZD prediction.

The analysis revealed that the drying duration parameter $(\boldsymbol{t_{dry}})$contributed the highest overall influence toward convection zone development. The wetting duration parameter $(\boldsymbol{t_{wet}})$exhibited the second-largest contribution, confirming the dominant role of cyclic wetting-drying exposure behavior identified during the symbolic regression analyses.

In comparison, the environmental variables $(\boldsymbol{q_{dry}})$and $(\boldsymbol{\theta_{r,eq}})$demonstrated comparatively smaller contributions toward the prediction process. These findings strongly supported the reduced-order normalized equation framework developed in the earlier sections.

The extracted feature importance distribution obtained from the Random Forest model is presented in Figure 4.3.

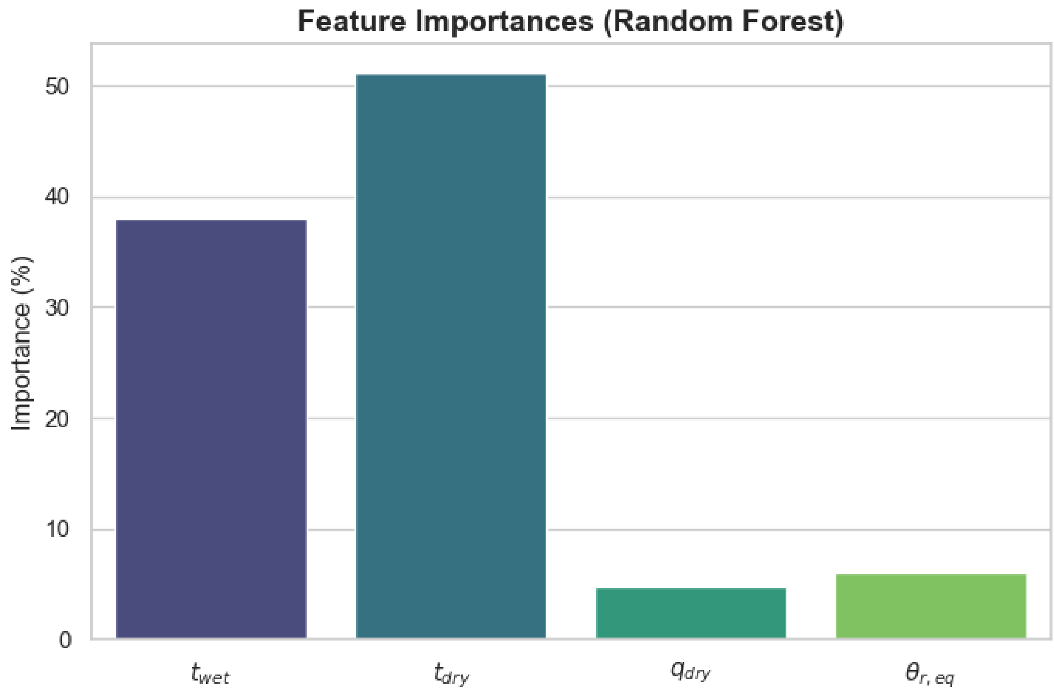

Figure 4.3 Random Forest Feature Importance Analysis of Environmental Variables Governing CZD Evolution

4.4.2 Validation of Dominant Temporal Variables

The Random Forest validation independently corroborated the conclusions obtained from the symbolic regression and DWTR-based coefficient analyses.

The temporal exposure variables (t_{dry}, t_{wet})consistently emerged as the governing parameters controlling CZD evolution, while the environmental weather-intensity parameters primarily acted as secondary modifiers of the transport response.

These observations confirmed that the convection zone formation process is predominantly governed by the cyclic interaction between wetting and drying exposure durations. The agreement between the symbolic regression framework, coefficient-based analyses, and Random Forest validation therefore demonstrated the robustness of the proposed reduced-order transport modeling approach under cyclic marine exposure conditions.

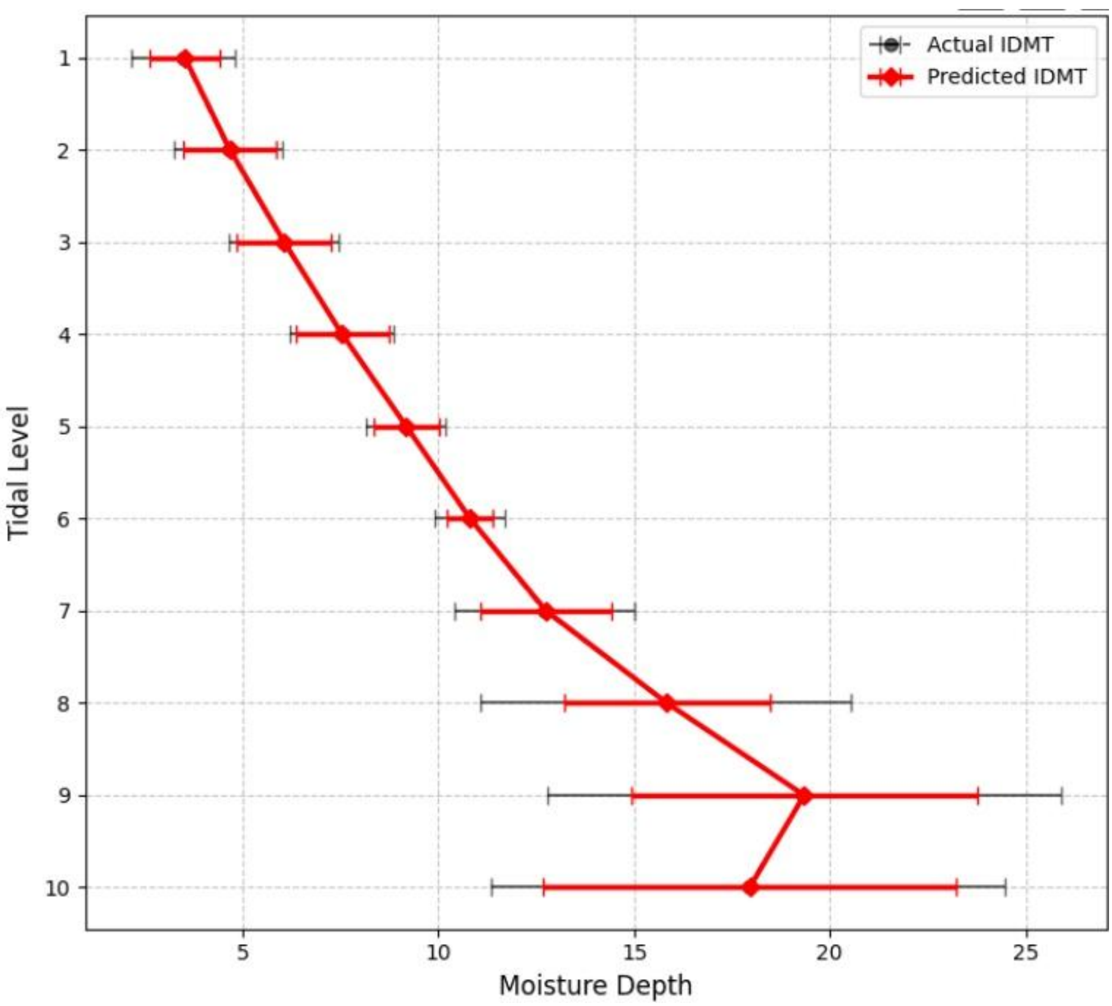

Figure 4.4 Comparison of Actual and Predicted IDMT Profiles Across Tidal Exposure Levels

The combined influence of drying-controlled and wetting-controlled transport mechanisms across the exposure profile is further illustrated in Figure 4.4. The observed IDMT profile demonstrates a progressive increase in convection-zone depth from the upper exposure levels toward the intermediate tidal region, followed by localized stabilization near the submerged levels. The horizontal black variance bars represent the variability range of the FEM-generated IDMT values at each tidal level.

The predicted profile generated using the proposed DWTR-based coefficient framework showed strong agreement with the FEM-derived profile distribution across the tidal levels. The agreement confirms that the reduced-order formulation successfully captures the dominant moisture transport evolution throughout the marine exposure cycle.

The profile behavior additionally supports the physical interpretation established through the coefficient analyses. Upper exposure levels exhibited comparatively lower moisture penetration associated with dominant drying-controlled transport behavior, whereas intermediate and lower exposure levels demonstrated increased wetting influence and deeper convection-zone development.

4.5 Discussion of Model Strengths and Limitations

The results of this study highlight the significant potential, as well as the inherent challenges, of utilizing a hybrid data-driven framework for predictive modeling in civil engineering. Transitioning from pure Symbolic Regression (SR) to strict parameter isolation and Machine Learning validation provided a comprehensive understanding of moisture transport dynamics.

4.5.1 Strengths of the Multi-Phase Approach

The proposed analytical framework demonstrated strong capability in identifying physically interpretable transport relationships governing convection-zone evolution under cyclic marine exposure conditions. A major strength of the framework lies in the integration of symbolic regression with DWTR-based coefficient analysis, enabling both mathematical interpretability and generalized trend identification.

Unlike purely black-box machine learning approaches, the symbolic regression stage generated explicit mathematical relationships directly linked to physically meaningful exposure variables. The repeated occurrence of square-root temporal relationships across independently trained datasets confirmed that the framework consistently captured dominant cyclic transport behavior rather than arbitrary numerical correlations.

The normalized reduced-order formulation further improved the simplicity and transferability of the analysis. By reducing the transport behavior into two governing coefficients (k_1, k_2), the framework successfully represented drying-controlled and wetting-controlled transport mechanisms while significantly reducing equation complexity.

The DWTR-based coefficient analysis additionally revealed systematic nonlinear transitions between drying-dominated and wetting-dominated exposure regimes. The generalized coefficient laws successfully reproduced the dominant coefficient trends observed across multiple coastal stations, suggesting the presence of universal transport behavior governing marine exposure conditions.

The independent validation performed on unseen coastal stations further demonstrated the robustness of the proposed framework. Figure 4.5 shows that the universal coefficient relationships were capable of reproducing the dominant IDMT profile evolution across stations S6–S10 with reasonable agreement. In several stations, the generalized framework successfully captured both the overall profile behavior and the dominant peak regions associated with maximum convection-zone development.

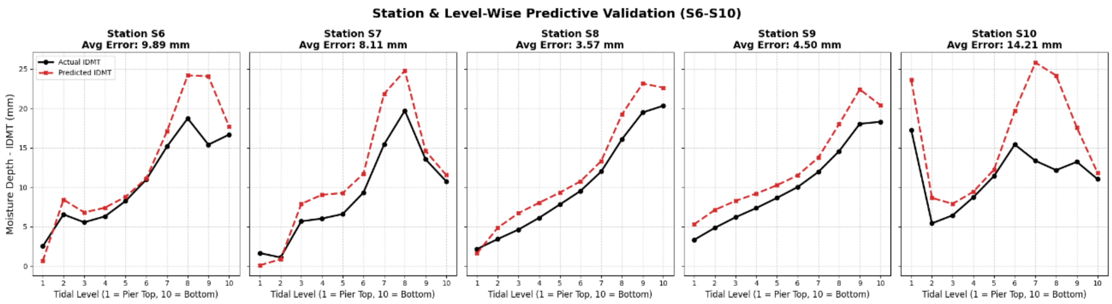

Figure 4.5 Independent Validation of the Universal DWTR-Based Model for Unseen Coastal Stations (S6–S10)

The framework also provided strong physical interpretability by directly linking cyclic exposure characteristics with moisture transport evolution. The observed transition between drying-controlled and wetting-controlled transport regimes remained consistently aligned with both the symbolic regression equations and the DWTR-based coefficient trends.

4.5.2 Limitations of the Analytical Framework

Despite the successful identification of dominant transport trends, several limitations were observed within the present analytical framework.

Although the generalized coefficient laws reproduced the dominant convection-zone evolution behavior, localized deviations remained present in certain exposure regions. Peak mismatches observed in stations such as S8–S10 indicate that station-specific environmental variability and localized numerical effects cannot be fully represented using a simplified two-parameter reduced-order formulation.

The symbolic regression framework additionally exhibited sensitivity to abrupt profile transitions presents within the FEM-generated datasets. Since symbolic regression fundamentally attempts to identify continuous analytical relationships, localized mesh-dependent profile discontinuities occasionally reduced predictive consistency for specific exposure levels.

The DWTR-based generalized coefficient relationships captured the dominant cyclic transport trends; however, secondary environmental influences were represented only indirectly through the extracted coefficients. As a result, localized environmental effects associated with complex marine exposure conditions may not be fully captured within the simplified framework.

Furthermore, the Random Forest validation demonstrated that temporal exposure variables dominated the convection-zone evolution process, while environmental weather-intensity variables primarily acted as secondary modifiers. Although this supported the reduced-order transport formulation, it also suggests that future studies may require additional environmental descriptors and higher-resolution datasets to improve localized predictive capability.

Overall, the multi-stage analytical work demonstrated that convection-zone evolution is primarily governed by cyclic temporal exposure behavior. The integration of symbolic regression, coefficient-based reduction, and machine learning validation enabled the extraction of physically interpretable transport relationships capable of reproducing dominant moisture transport trends across varying marine exposure conditions

Chapter 5

Conclusions and Future Work

This chapter summarizes the major findings of the present study and highlights the future research directions emerging from the developed analytical framework. The study successfully demonstrated the feasibility of integrating Symbolic Regression (SR), reduced-order coefficient modelling, and Machine Learning validation for interpreting and predicting convection-zone evolution under cyclic marine exposure conditions. The proposed framework not only provided accurate predictive capability, but also established physically interpretable relationships governing moisture transport behavior in marine concrete environments.

5.1 Conclusion

The primary objective of this research was to investigate the governing transport mechanisms responsible for Convection Zone Depth (CZD) evolution under cyclic wetting-drying exposure and to develop a generalized predictive framework based on physically meaningful environmental parameters. The major conclusions obtained from this study are summarized as follows:

1. Symbolic Regression Successfully Identified Dominant Physical Relationships: The PySR Symbolic Regression framework successfully generated compact mathematical expressions linking CZD evolution with cyclic exposure variables. The discovered symbolic equations repeatedly incorporated temporal exposure terms involving drying duration $(t_{dry})$and wetting duration (t_{wet}), particularly through nonlinear square-root relationships.

The repeated emergence of these structures across independently trained exposure levels indicated that the symbolic regression process was successfully identifying stable physical transport behavior rather than arbitrary numerical correlations.

2. Temporal Exposure Variables Govern Convection-Zone Evolution: The analysis demonstrated that the drying and wetting durations are the dominant governing parameters controlling CZD development under marine exposure conditions.

Both the DWTR-based coefficient analysis and Random Forest validation consistently showed that:

- drying duration (t_{dry}) contributes the largest influence toward convection-zone evolution,
- wetting duration (t_{wet})acts as the secondary governing parameter,
- environmental weather-intensity variables $(q_{dry}, \theta_{r,eq})$ contribute comparatively smaller effects.

This confirms that convection-zone formation is primarily controlled by the cyclic interaction between drying and wetting exposure durations.

3. Reduced-Order Coefficient Modelling Successfully Captured Transport Behavior: The normalized reduced-order formulation:

$$CZD = k_1\sqrt{t_{dry}} + k_2\sqrt{t_{wet}}$$

successfully represented the dominant transport behavior while significantly reducing equation complexity. The extracted transport coefficients revealed systematic nonlinear transitions between drying-dominated and wetting-dominated exposure regimes throughout the tidal profile.

The drying-related coefficient $k_1$demonstrated a monotonic decay trend with increasing Drying-to-Wetting Time Ratio (DWTR), whereas the wetting-related coefficient $k_2$exhibited a nonlinear peak-type response across intermediate DWTR values.

4. Universal DWTR-Based Coefficient Relationships Were Established: The DWTR-based analysis enabled the development of generalized analytical relationships representing the evolution of the transport coefficients across multiple coastal stations and exposure levels.

The generalized coefficient functions successfully reproduced the dominant station-wise trends and demonstrated that convection-zone transport behavior transitions progressively between drying-controlled and wetting-controlled regimes depending on the cyclic exposure ratio.

These universal coefficient relationships provide a physically interpretable framework for estimating dominant moisture transport behavior directly from exposure-cycle characteristics.

5. Machine Learning Validation Independently Confirmed the Physical Trends: The Random Forest validation framework independently corroborated the conclusions obtained from the symbolic regression and DWTR-based analyses.

Feature-importance analysis demonstrated that the temporal exposure variables dominated CZD prediction. Environmental weather-intensity parameters acted mainly as secondary modifiers. The validation therefore confirmed the robustness of the reduced-order transport framework and supported the physical interpretation established through the coefficient analysis.

6. The Proposed Framework Demonstrated Strong Generalization Capability: Independent validation performed on unseen coastal stations demonstrated that the generalized DWTR-based coefficient framework was capable of reproducing the dominant IDMT profile behavior with reasonable agreement.

Although localized deviations remained present for certain stations and exposure levels, the framework consistently captured:

- the overall profile evolution,
- dominant peak regions,
- and the transition between drying-controlled and wetting-controlled transport mechanisms.

This demonstrates the capability of the proposed framework to generalize dominant marine moisture transport behavior across varying exposure environments. Overall, the present study established that convection-zone evolution under marine exposure conditions can be effectively represented using a simplified physically interpretable transport framework governed primarily by cyclic wetting-drying exposure durations.

5.2 Future Work

Although the proposed framework successfully captured the dominant transport behavior governing CZD evolution, several opportunities remain for improving the robustness, generalization capability, and engineering applicability of the methodology.

The following directions are recommended for future research:

1. Expansion to Larger Multi-Station Datasets: The present study validated the framework across selected coastal stations and exposure levels. Future work should extend the methodology to larger multi-regional marine datasets containing broader climatic, tidal, and environmental variability.

Validation across a wider range of coastal environments would help establish the universal applicability of the proposed DWTR-based transport relationships.

2. Development of a Fully Unified Predictive Equation: Although generalized coefficient functions were successfully derived, the framework still relies on intermediate coefficient estimation procedures.

Future studies may focus on developing a single fully unified analytical equation capable of directly predicting CZD from environmental exposure variables while maintaining both physical interpretability and predictive robustness.

Constrained Symbolic Regression approaches guided by the identified dominant temporal variables may provide a promising direction for achieving this objective.

3. Inclusion of Additional Environmental Parameters: The present reduced-order framework primarily considered cyclic wetting-drying exposure behavior. However, marine durability processes may also be influenced by additional environmental factors such as:

- chloride concentration,
- temperature fluctuations,
- relative humidity,
- wind intensity,
- and seasonal exposure variability.

Incorporating these parameters may improve predictive capability for highly heterogeneous marine exposure environments.

4. Integration with Long-Term Durability and Service-Life Models: Future work may integrate the proposed convection-zone transport framework with long-term chloride ingress and corrosion-propagation models.

Such integration could support the development of practical service-life prediction tools for reinforced concrete structures.

5. Improvement of Localized Prediction Capability: The validation results revealed that localized profile deviations and abrupt mesh-dependent transitions occasionally reduced predictive consistency at certain exposure levels.

Future studies may investigate:

- higher-resolution FEM datasets,
- adaptive regional coefficient fitting,
- or hybrid physics-informed machine learning frameworks to better capture localized transport irregularities while preserving physical interpretability.

6. Application to Real Experimental and Field Data: The present study primarily utilized FEM-generated exposure datasets. Future validation using laboratory experiments and real marine field measurements would provide stronger confirmation of the physical applicability of the developed framework.

Experimental calibration could also help refine the universal coefficient relationships for practical engineering implementation.

In summary, the present work establishes a strong foundation for interpretable data-driven modelling of marine convection-zone evolution and provides a pathway toward future development of generalized predictive durability frameworks for coastal concrete infrastructure.

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Appendix

Coefficients Table:

Station	Level	k1	k2	R2	RMSE
S1	1	3.20	-0.002	0.58	0.82
S1	2	3.72	-0.02	0.94	0.29
S1	3	3.94	-0.07	0.95	0.28
S1	4	4.16	-0.15	0.95	0.26
S1	5	4.29	-0.20	0.91	0.27
S1	6	3.74	0.44	0.74	0.40
S1	7	3.62	0.91	0.80	0.98
S1	8	2.03	4.67	0.45	3.16
S1	9	0.38	11.58	0.46	4.07
S1	10	0.04	14.41	0.85	1.46
S2	1	3.09	-0.01	0.78	0.58
S2	2	2.30	0.13	-0.21	3.00
S2	3	2.43	0.49	0.11	3.87
S2	4	2.50	0.84	0.24	2.95
S2	5	2.36	1.60	0.39	2.78
S2	6	1.83	3.33	0.33	4.37
S2	7	0.89	7.22	0.31	9.31
S2	8	0.72	5.32	-0.34	10.64
S2	9	0.20	4.87	-0.42	10.64
S2	10	0.07	6.63	-0.14	6.35
S3	1	2.72	-0.01	0.67	0.87
S3	2	3.17	-0.04	0.94	0.44
S3	3	3.45	-0.11	0.95	0.48
S3	4	3.31	-0.04	0.71	1.28
S3	5	2.93	0.32	0.52	1.75
S3	6	3.17	0.46	0.62	2.22
S3	7	1.82	3.88	0.19	5.07
S3	8	0.34	10.98	0.10	8.12
S3	9	-0.02	16.12	0.60	6.93
S3	10	-0.07	19.8988	0.7828	5.2166

S4	1	1.46	0.01	0	0.10
S4	2	4.27	-0.01	0.99	9.97
S4	3	4.27	-0.01	0.99	4.05
S4	4	4.27	-0.04	0.99	1.64
S4	5	3.35	0.06	0.96	1.96
S4	6	1.41	1.31	-0.33	3.83
S4	7	1.03	3.01	-0.41	4.72
S4	8	0.75	5.66	-0.11	6.35
S4	9	0.30	9.07	-0.11	6.57
S4	10	0.01	15.13	0.59	4.44
S5	1	3.42	-0.0	0.75	0.73
S5	2	3.78	-0.02	0.95	0.31
S5	3	3.99	-0.07	0.96	0.31
S5	4	4.24	-0.21	0.96	0.26
S5	5	4.18	-0.09	0.82	0.56
S5	6	3.23	1.41	0.40	2.19
S5	7	1.54	5.56	0.15	4.55
S5	8	0.44	10.85	0.38	4.10
S5	9	0.02	14.75	0.73	2.48
S5	10	0	15.20	0.34	0